

Agent-Infrastructure Fit: How AI Agents Are Redefining the Governance of Public Digital Data Infrastructure

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Abstract

As AI agents enter civic analytics, public-sector decision-making, and policy research, the infrastructure quality of government open-data portals becomes a critical governance problem. Existing evaluation frameworks emphasize legal availability, metadata completeness, and human-facing demand, but rarely test whether AI agents can successfully use public data. This paper introduces agent-infrastructure fit: the alignment between task requirements and the affordances a public-data system provides. We argue that agentic use expands open-data governance beyond publication: datasets must also be queryable, joinable, semantically documented, and resilient to machine failure modes. The evidence has three layers. First, Socrata/Tyler portal logs from San Francisco, New York City, and Seattle from 2020 to 2026 provide secondary associational context. High-friction datasets exhibit stronger post-2025 API/download growth relative to page views, most clearly in New York City; the design does not identify a causal AI effect. Second, we develop an Agent-Readable Infrastructure Index (ARI) covering six dimensions: metadata completeness, schema documentation, API queryability, joinability, semantic/reference support, and operational readiness. Third, a controlled benchmark executes 90 civic-data tasks three times under no-metadata, schema-only, and structured-ARI conditions, producing 810 completed runs. Schema access achieves the highest automatic task-success rate (62.2%), compared with structured ARI cards (52.2%) and no metadata (43.0%). Structured ARI cards nevertheless reduce hallucinated fields by 89.5% relative to schema access and reduce wrong joins from 0.15 to 0.02 per run. Automatic validators flag missing reference assets in 59.1% of runs. These results reveal a performance-reliability frontier: concise schemas support completion, while structured semantic constraints reduce critical errors. The paper shifts open-data evaluation from static availability toward relational, task-level agent usability and provides a governance diagnostic for auditing whether public digital infrastructure is ready for AI-mediated analytical work.

Keywords: AI agents; agent-infrastructure fit; open government data; agent-readable infrastructure; digital public infrastructure; data governance